

Carolyn Reichherzer

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Research Profile: Across academic and industry roles, I have designed, developed, and evaluated how immersive technology shapes perception, decision-making, and user behaviour, with the intention of better understanding novel technologies and improving how we engage with them. I am now leveraging this expertise to investigate how XR combined with AI can shape and augment user trust and behaviour in human-AI interaction, with a particular focus on transparent and reliable decision support. With the rise of more accessible form factors in immersive technologies, better sensors, and the integration of AI, I aim to be a pioneer in developing new ways in which these technologies can be used responsibly and effectively within civil engineering.

Education

2021 - PhD in Computer Science | University of South Australia (now Adelaide University)

Master of Science in Digital Media | University of Bremen

Bachelor of Arts in Communication Design | University of Applied Sciences Mainz

Experience

Postdoctoral Fellow | ETH Zürich

July 2026 – Present

- Investigating the use of Augmented Reality visualisation techniques and LLM traits to enhance collaboration, trust and appropriate reliance for effective decision-making.
- Developing novel interaction paradigms for a new generation of lightweight XR headsets in combination with conversational agents.
- Co-Leading the Immersive Design Lab, providing XR expertise and resources to AEC researchers at ETH.

Senior Research Associate | Lucerne University of Applied Sciences and Arts

Nov 2024 – June 2026

- Led mixed-methods studies on photorealistic immersive experiences with novel 3D representation, evaluating perception and user impact to inform design for immersive experiences, and conducted statistical analyses.
- Initiated research on how LLM personality traits, context and embodiment influence user trust, with the goal of informing conversational AI design practices.
- Contributed to experimental design for a project on conversational agents for spiritual guidance.

UX Researcher | Meta

March 2023 – September 2024

- Evaluated the avatar motion quality of AI-driven tracking technology in Virtual Reality via lab experiments.
- Led foundational and generative research on the quality of photorealistic avatars.
- Under uncertainty, steered strategy and direction based on study outcomes, influenced product roadmaps, and communicated actionable research findings to various stakeholders clearly and concisely.
- Built and managed the first User Experience Research lab in Zurich to drive research initiatives in this location.

Senior Research Associate | Lucerne University of Applied Sciences and Arts

April 2021 – March 2023

- Facilitated evaluative and generative research on usability and perception for various AR/VR applications via lab experiments, surveys, interviews, and logged user behaviour. Conducted qualitative and quantitative analyses.
- Project lead of cross-disciplinary projects with external stakeholders to align on needs and priorities, and presented research insights to both academic and industry partners.

Research Associate | Australian Research Centre for Interactive and Virtual Environments

October 2020 – March 2021

- Led investigations and co-developed novel cross-device user experiences, specifically exploring the simultaneous interaction between head-mounted displays and mobile devices to assess how we can better work with AR.
- Evaluated diverse user interfaces through rigorous qualitative and quantitative user studies, assessing UX quality and the impact of critical factors like field of view on user performance and perception.

PhD Candidate | Australian Research Centre for Interactive and Virtual Environments

December 2016 – March 2021

- Effectively collaborated with a diverse group of law professionals and scientists via workshops and expert interviews to set the foundation for utilising VR for courtroom applications to improve decision-making and ease mental effort on trial participants.
- Developed prototypes in Unity and conducted user research using interviews, surveys, behavioural logging, eye-tracking, and physiological sensing to assess cognitive performance in immersive environments.

- Mentored students and facilitated undergraduate programming tutorials.

Research Assistant | University of South Australia (now Adelaide University)

October 2015 - April 2016

- Developed and evaluated hand-tracking interaction techniques for improved navigation of large datasets in VR using Leap Motion and Unity.

Research Assistant | Human Interface Technology Laboratory New Zealand

July 2013 – June 2014

- Developed an interactive prototype to explore creative solutions for sharing experiences remotely via graphical annotation with Google Glass.
- Planned, executed, and analysed user studies using subjective and objective measures for remote social connection with Google Glass.

Awards & Achievements

2022 **Best Paper Nomination (Top 20)**: IEEE VR 2022.

2021 **Honourable Mention – Best of CHI (Top 5%)**: CHI Conference 2021.

2020 **Research Excellence Awards Nominee**: University-wide nominee for STEM. University of South Australia.

2019 **1st Place Retail Game Changer Challenge**: \$25 000 AUD for winning prototype, Precision Group, Australia.

2019 **Highly Commended**: „Academic Excellence – Postgraduate Research“, South Australian government.

2019 **Shortlisted**: **Google Women Techmaker APAC** Finalist for the Asia-Pacific region (20 000+ applicants), Google.

Academic Service

since 2021 **Committee Member** for Conference on Human Factors in Computing Systems (CHI)

Reviewer for IEEE VR, ACM CHI, IEEE ISMAR, ACM TOCHI, PLOS, Journal of Human-Computer Interaction

Student Supervision

Master student (active): 1, Bachelor student (past): 1. Mentored 1 PhD student over 2 years with meetings 2x/month and 2 publications.

Community

2018 - 2021 **Women in STEM UniSA**: Co-founder and committee member, we won multiple awards for our engagement with students.

2017 - 2021 **Code Like a Girl**: Event volunteer and panellist, supporting the WSTEM community in Adelaide.

2019 **TechStars Startup Weekend Adelaide**: Tech Mentor.

Publications (most important)

[Honourable Mention] Reichherzer C. et al., Bringing the Jury to the Scene of the Crime: Memory and Decision-Making in a Simulated Crime Scene. In CHI Conference on Human Factors in Computing Systems (CHI '21), May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 12 pages, 2021.

[Best Paper Nomination] Reichherzer C. et al., „Supporting Jury Understanding of Expert Evidence in a Virtual Environment,“ 2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2022, pp. 615-624, doi: 10.1109/VR51125.2022.00082.

[Selected for special issue journal publication] Reichherzer C. et. al, „Narrative and Spatial Memory for Jury Viewings in a Reconstructed Virtual Environment“ in IEEE Transactions on Visualization and Computer Graphics, 2018.

Reichherzer C. et. al., 2021. SecondSight: Demonstrating Cross-Device Augmented Reality. In ACM SIGGRAPH 2021 Immersive Pavilion (SIGGRAPH '21). Association for Computing Machinery, New York, NY, USA, Article 6, 1–2. <https://doi.org/10.1145/3450615.3464545>

Publications (most relevant)

Haslbauer, P., **Reichherzer, C.**, & Smolic, A. Deus in Machina: A Conversational AI for Spiritual Experience, International Conference on Digital and Interactive Arts, 2025.

[To Appear] Reichherzer C. et al., „User Perceptions of Gaussian Splats for Immersive Experiences,“ Quality and User Experience, Springer.

Reichherzer C. et al. 2021. SecondSight: A Framework for Cross-Device Augmented Reality Interfaces. In CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts), May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages, 2021.

Matthews, B. J., **Reichherzer, C.**, & Smith, R. T. (2023, April). Remapped interfaces: Building contextually adaptive user interfaces with haptic retargeting. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (pp. 1-4).

Haslbauer, P., Pullen, M., Praprotnik, M., Stimec, S. P., Grosbois, A., **Reichherzer, C.**, & Smolic, A. (2025, June). VR Planica: Gaussian Splatting Workflows for Immersive Storytelling. In ACM International Conference on Interactive Media Experiences Workshops (IMXw) (pp. 47-53). SBC.